

# 400 Gbit/s IM/DD PON with >30 dB Power Budget Enabled by Quantum-Dot SOA Amplified Remote Node

Ahmed Galib Reza<sup>(1)</sup>, Janina Rautert<sup>(2)</sup>, Vladimir S. Mikhlin<sup>(2)</sup>, Alexey Gubenko<sup>(2)</sup>, and Liam P. Barry<sup>(1)</sup>

<sup>(1)</sup> School of Electronic Engineering, Dublin City University, Dublin 9, Ireland, [ahmed.galibreza@dcu.ie](mailto:ahmed.galibreza@dcu.ie)

<sup>(2)</sup> Innolume, Konrad-Adenauer-Allee 11, 44263 Dortmund, Germany

**Abstract** We demonstrate 4×100 Gbit/s NRZ-OOK transmissions with 200 GHz channel spacing for next-generation 400G PON. By employing an InAs/GaAs quantum-dot SOA-amplified remote node, we achieve an optical power budget beyond N1 class after 20 km at the HD-LDPC BER limit, using a T-spaced FFE. ©2026 The Author(s)

## Introduction

The passive optical network (PON) is the most widely used technology to offer high-speed broadband services to fiber-to-the-home (FTTH) users. The PON infrastructure now connects over 65% of the FTTH deployments, supporting more than 700 million users globally [1]. The latest PON standard, known as 50G-PON defined in the ITU-T G.9804.3 or HSP (Higher Speed PON) [2], supports 50 Gbit/s downstream and 12.5/25/50 Gbit/s upstream transmissions over a single wavelength in the O-band using high-power lasers, digital signal processing (DSP), and an enhanced forward error correction (FEC) limit, reaching five times more capacity than previous generation XGS-PON [3].

The ITU-T is currently working on technologies beyond 50 Gbit/s PON called Very High Speed PON (VHSP), anticipating a capacity of at least 200 Gbit/s [4]. To overcome the bandwidth and dispersion factors in achieving 200G/λ [5], some recent demonstrations consider utilizing coherent solutions for PONs, such as simplified coherent [6,7] and self-coherent [8] systems. As the cost is still a decisive factor for PONs from deployment to operation, and all previous standards are based on intensity modulation with direct detection (IM/DD) technology, several 200G IM/DD PON solutions have also been reported based on a single wavelength 200 Gbps system utilizing complex DSP [9], and costly components such as Raman amplifiers [10] and Praseodymium-doped fiber amplifiers (PDFA) [9]; and dual wavelengths with 100 Gbps/λ [11-13]. Recently, utilizing dual-wavelength and dual-polarization, a 400G IM/DD PON has been proposed and demonstrated [14], as 400G PON is also under consideration of the study group as VHSP within ITU-T SG15/Q2. One critical issue, especially with 400G PON, is satisfying the N1-class power budget requirement of a legacy PON. The total optical launch power into the fiber for 200G PONs is already approaching 15 dBm [6-13]. A further

increase in launch power to meet the optical signal-to-noise ratio (OSNR) requirement of the receiver for 400G data rates, using single or multiple channels, can lead to significant nonlinear fiber effects. These include stimulated Brillouin scattering (SBS), self-phase modulation (SPM), and four-wave mixing (FWM). Such effects degrade signal quality and become more of an issue with multi-wavelength systems.

To keep the signal launch power at an acceptable level to overcome fiber nonlinearity, this work proposes the application of an amplified remote node (RN) in a 400G wavelength-division multiplexed (WDM)-PON that concurrently amplifies all the operating wavelengths in the network to boost the power budget [15-17]. We validate that a semiconductor optical amplifier (SOA) based on quantum dot (QD) technology is a suitable choice in this regard, as it manifests low cross-gain modulation (XGM) and patterning effects at high speeds [17,18]. We experimentally demonstrate 4×100 Gbit/s NRZ-OOK transmission in the O-band utilizing a QD-SOA-amplified RN and report optical power budgets ranging from 30.7 dB to 34.7 dB after 20 km in a single-mode fiber (SMF). Such an SOA-amplified RN can be operated on a long-life battery, solar cell, or housed in a cabinet with power units.

## Experimental Setup

The experimental setup is illustrated in Fig. 1. The optical line terminal (OLT) utilizes four discrete distributed-feedback (DFB) lasers to achieve 4×100 Gbit/s transmission. To realize a ≈200 GHz-spaced WDM grid, the wavelengths of the lasers are tuned to 1309.58 nm (Ch-1), 1310.72 nm (Ch-2), 1311.86 nm (Ch-3), and 1313 nm (Ch-4) by adjusting the temperature of each device using a thermoelectric cooler (TEC). To study the impact of XGM and FWM during multi-wavelength amplification of high-speed signals in a QD-SOA, the uncorrelated data loading in neighboring channels is critical. It is achieved by combining the continuous wave (CW) outputs

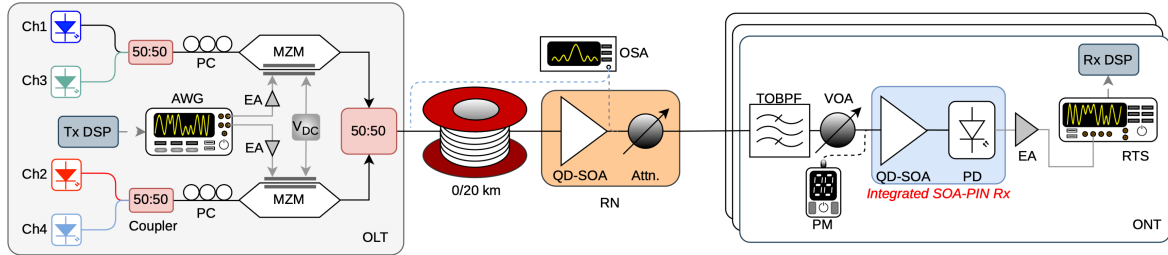


Fig. 1: Experimental setup for 4×100 Gbit/s NRZ-OOK transmission over 20 km.

from Ch-1 and Ch-3 (Group 1), and Ch-2 and Ch-4 (Group 2), using two 50:50 passive couplers and modulating with two independent Mach-Zehnder Modulators (MZMs) with a 6-dB bandwidth of 40 GHz, which are driven by two uncorrelated 100 Gbit/s NRZ-OOK signals. Finally, the 4×100 Gbit/s WDM signals are realized by combining the MZM outputs with a 3-dB coupler and launched over back-to-back (BtB) and 20 km SMF with an attenuation coefficient of 0.35 dB/km for transmission and performance analysis.

The two uncorrelated 100 Gbit/s NRZ-OOK signals are generated in MATLAB from a pseudo-random bit sequence (PRBS) with a length of  $2^{17}-1$ . The NRZ-OOK signals are upsampled twice to align with the sampling rate of 200 GSa/s of an arbitrary waveform generator (AWG), and a root-raised cosine (RRC) pulse-shaping with a roll-off factor of 0.1 is applied to reduce bandwidth. The digital NRZ-OOK bit streams are loaded into a two-channel AWG with 70 GHz analog bandwidth to produce uncorrelated 100 Gbit/s analog NRZ-OOK signals with a peak-to-peak voltage ( $V_{pp}$ ) of 325 mV. The AWG outputs are sent to two RF amplifiers with 3-dB bandwidths and the gains of 55 GHz and 23 dB, respectively, to drive the MZMs.

In the RN of the PON, a high-gain QD-SOA with a cavity length of 6 mm is used to compensate for the transmission and splitter losses in the network. The QD-SOA offers a noise figure (NF) of 5.6 dB when biased at 600 mA and operated at 25°C, with a gain peak located at 1304.2 nm and offering a 3-dB bandwidth of 18 nm. As the proposed 400G PON operates over a range of wavelengths near 1310 nm, the SOA gain spectrum enables WDM operation.

The optical network terminal (ONT) receiver optics comprise a tunable optical bandpass filter (TOBPF), an SOA pre-amplifier, and a photodetector (PD). First, the received signals are demultiplexed by tuning the bandwidth and wavelength of a TOBPF. To improve receiver sensitivity, a pre-amplified QD-SOA-PIN configured receiver design is adapted in this work, which offers potential for hybrid/monolithic integration on a photonic chip. This QD-SOA is operated at a fixed bias current of 700 mA and a temperature of 25°C, offering a low-noise pre-amplification with

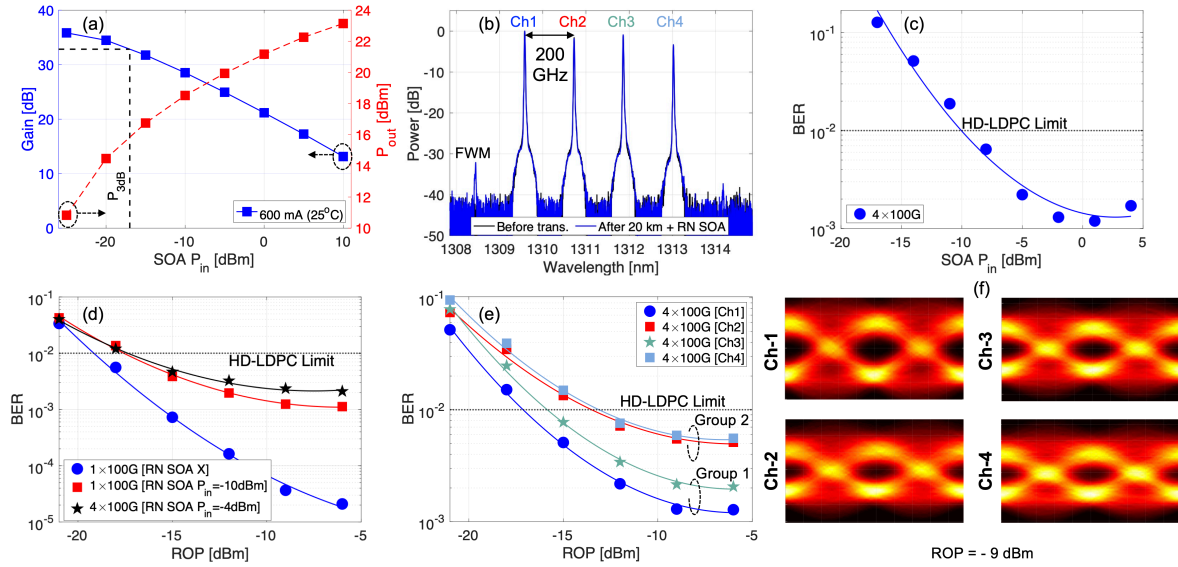
a typical gain of about 14 dB. Next, the demultiplexed signals are detected using a 70 GHz PD with a responsivity of 0.6 A/W, followed by amplification with a 66 GHz bandwidth RF amplifier and capturing by a 256 GSa/s real-time scope (RTS) with a 55 GHz brick-wall frequency response. The captured signals are processed offline, which comprises RRC matched filtering, resampling at 1 Sa/symbol, synchronization, and digital equalization using a feed-forward equalizer (FFE) with 32 taps to mitigate signal distortions. Finally, we measure the bit error ratio (BER) for performance evaluation.

### Experimental Results

In Fig. 2(a), the measured gain and output power of the QD-SOA operated at the RN are presented versus SOA input power for an input wavelength of 1310 nm at 600mA bias and 25°C. The SOA offers a large unsaturated gain of 35.83 dB, which validates its suitability in the RN to boost the power budget. Despite producing a large unsaturated gain, the 3-dB saturation input power for the SOA is -17 dBm (at which point the SOA gain is reduced by 3 dB from the unsaturated point). Therefore, the QD-SOA amplified signal may experience some level of nonlinear distortions, which can degrade signal quality.

Fig. 2(b) shows 100 Gbit/s NRZ-OOK modulated four-wavelength WDM spectra with 200 GHz spacing before launching (black), and after 20 km transmissions over an SOA amplified RN (blue). In the experiment, the signal launch power into the fiber is kept as low as 4 dBm (-2 dBm/λ) to ensure there is no FWM in the fiber. However, due to injecting an input power of -4 dBm and four narrowly spaced wavelengths into the SOA, some level of FWM occurs in the SOA.

Next, in Fig. 2(c), to assess the dynamic range of the QD-SOA running in the RN during 4-wavelength amplifications, we measure the BERs of Ch-1 (1309.58 nm) versus total input power into the RN SOA for a BtB case with a received optical power (ROP) of -9 dBm. Throughout the experiment, the ROP is measured at the input to the integrated SOA-PIN receiver. It is found that the hard-decision low-density parity-check (HD-LDPC) level BER can be reached employing an RN SOA with input powers ranging from -10 dBm



**Fig. 2:** (a) Gain and output power of the QD-SOA versus input power. (b) Modulated spectra before launching (black) and after transmissions over 20 km with SOA amplification in the RN (blue). (c) BER of Ch-1 after 4x100 Gbit/s BtB transmission versus input power into the RN SOA. (d) BER versus ROP of Ch-1 after BtB transmissions over an unamplified and amplified RN. (e) BER versus ROP of all channels after 20 km over an amplified RN. (f) Equalized eye diagrams of all four WDM channels after 20 km transmissions.

to 4 dBm. At an input power of 4 dBm, distortion is observed due to SOA nonlinearities. At a low SOA input power of <-10 dBm, the SOA-induced ASE noise dictates the system performance.

In Fig. 2(d), we plot the BER results of Ch-1 after BtB transmission versus ROP for three transmission scenarios: (1) unamplified RN with single wavelength, (2) amplified RN with single wavelength, and (3) amplified RN with 4x100 Gbit/s transmission. Considering the transmission loss of a 20 km SMF, the total SOA input power is set to -4 dBm (-10 dBm/λ) with a VOA. As expected, the unamplified RN offers superior performance in contrast to the amplified RN. However, the performance disparity between the single- and four-wavelength amplification scenarios is comparable at the HD-LDPC level BER threshold, and it shows about a 1.7dB penalty compared with the unamplified case. The QD-SOA is thus suitable for deployment in an RN with minimal signal distortions during multi-wavelength amplifications for 4x100G WDM-PON.

The BERs of all four channels are presented in Fig. 2(e) after 20 km transmission using the QD-SOA amplified RN. We can see that all WDM channels can guarantee an HD-LDPC level performance at an ROP ≤ -13.1 dBm. Considering the launch power of 4 dBm, the SOA gain of 24.1 dB for an input power of -4 dBm, and a 4.5 dB insertion loss for the TOBF, the achievable optical power budget after 20 km ranges from 30.7 dB to 34.7 dB (channel dependent), at the HD-LDPC BER limit of 10<sup>-2</sup>. The performance disparity between Group 1 and Group 2 is due to distinct modulator responses and polarization sensitivity of the QD-SOA. Through precise

polarization adjustment or by designing a polarization-insensitive SOA [19], the performance disparity can be minimized, enabling all channels to reach ~35 dB power budget for a launch power that does not trigger fiber nonlinearities.

Finally, in Fig. 2(f), the equalized eye diagrams of all four channels after 4x100 Gbit/s transmission over 20 km are plotted at an ROP of -9 dBm. The best eye openings are observed for Ch-1, followed by Ch-3, Ch-2, and Ch-4, which confirms the BER performances obtained in Fig. 2(e). The clear eye openings for all channels with an FFE demonstrate that linear distortion caused by bandwidth limitation of the system is the major source of impairment to achieve 100 Gbit/s/λ data rates, and nonlinear signal distortions caused by a QD-SOA, such as XGM, FWM, are negligible.

## Conclusions

We report 4x100 Gbit/s NRZ-OOK transmissions in the O-band for next-generation 400G PON. By employing a QD-SOA amplified RN, we show the feasibility of realizing 400G PON using a simple DSP and low-cost IM/DD technology with negligible nonlinear signal distortions and an improved power budget. Depending on the channel, we report optical power budgets varying from 30.7 dB to 34.7 dB at the HD-LDPC BER limit after transmission over 20 km SMF for a total launch power of only 4 dBm, ensuring no fiber nonlinearities during transmission. In summary, the proposed SOA-amplified RN architecture can be a viable option to facilitate the next-generation 400G VHSP employing WDM based on a simpler IM/DD technology.

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